



## Program 1: Vector Addition ( $A + B = C$ )

### Description

This program performs a vector addition on two vectors A and B, followed by a vector clipping operation. Vector A is (0, 0, 2, 5) and Vector B is (0, 0, 3, 4). The resulting vector (0, 0, 5, 9) is bounded by a maximum value of 15. The final output is displayed on the LEDs.

### Register Map:

- R1 = Pointer to Vector A
- R2 = Pointer to Vector B
- R3 = Resulting Vector C
- R4 = Clipped Vector C
- R31 = Clip Limit (15)

| Assembly Code |             | Binary (Hex) |
|---------------|-------------|--------------|
| LOAD          | R1, 0(R0)   | 1C010000     |
| LOAD          | R2, 1(R0)   | 1C020001     |
| VADD          | R3, R1, R2  | 40221800     |
| ADDI          | R31, R0, 15 | 241F000F     |
| VCLIP         | R4, R3, R31 | C0802780     |
| OUT           | R4          | 34800000     |

